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15 December 1981
SUPERSEDING
MIL-L-81913(AS)
5 January 1973

MILITARY SPECIFICATION

LEATHER, CATTLEHIDE, VEGETABLE TANNED, FOR PACKING AND SEALS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of vegetable tanned cattlehide leather.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/4	- Aluminum Alloy 2024, Plate and Sheet.
QQ-B-626	- Brass, Leaded and Nonleaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-750	- Bronze, Phosphor; Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.

MILITARY

MIL-S-7720	- Steel, Corrosion-Resistant (18-8) Bars, Wire and Forging Stock (Aircraft Quality).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8330

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MIL-S-18729 - Steel Plate, Sheet, and Strip, Alloy, 4130, Aircraft Quality.
MIL-L-43432 - Leather, Packaging and Packing of.

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of conflict between this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. The leather shall be made from prime quality cattlehide, containing no holes or brands in the bend area.

3.2 Tannage. The leather shall be vegetable tanned.

3.3 Impregnation. The leather shall be impregnated with a mixture of oils and greases or other compounds applicable to conditions of intended use.

3.4 Physical requirements.

3.4.1 Trim. The leather shall be furnished in the form of bends and shall be trimmed to conform with the definition of bends as listed under "Definition of Terms" in FED-STD-311. The area in terms of square feet shall be legibly marked in the butt area of the flesh side of the leather.

3.4.2 Thickness. The thickness of the leather shall be 8 oz. ± 1 oz. NOTE: 1 ounce = 1/64 inch.

3.4.3 Finish. The leather shall be undyed with no surface finish.

* 3.4.4 Corrosive effect. The leather shall be non-corrosive and shall show no signs of dulling the polished surfaces when tested as specified in 4.3.

3.4.5 Resistance to low temperature. The leather shall remain pliable and flexible when tested as specified in 4.3.1.

3.4.6 Resistance to high temperature. The leather shall not crack when tested as specified in 4.3.2.

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* 3.4.7 Tensile strength. The leather parallel with the backbone shall have a tensile strength of 4500 pounds per square inch and the leather perpendicular to the backbone shall have a tensile strength of 3000 pounds per square inch when tested as specified in 4.3.

3.4.8 Chemical requirements. The chemical requirements of the leather shall be as specified in table I when tested as specified in 4.3.

TABLE I. Chemical properties.

Characteristic	Requirement
Total ash, percent, max 1/	0.35
Water soluble material, percent, max 1/	12.1

1/ On fat free basis

3.5 Workmanship. The leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as quality conformance inspection. The qualify conformance inspection shall consist of all the examinations and tests required by this specification. Sampling for inspection shall be performed in accordance with FED-STD-311, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable acquisition documents.

4.2.2 Examination of the end item. The leather shall be examined for the defects listed in 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one bend. The lot size shall be expressed in units of one bend. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with

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FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more units of product. Separate examinations shall be performed for visual and dimensional characteristics, but the same sample may be used for each examination.

4.2.2.1 Visual examination.

Examine	Defect
Material	Not as specified.
Trim	Not a bend.
Color	Color not natural as produced by tanning process.
Finish	Not as specified.
Quality of leather	Cut, scratch, hole or brand.

NOTE: Light or well healed scratches, well healed grub damage, light fat wrinkles or slight stains shall not be classified as defects.

4.2.2.2 Dimensional examination.

Examine	Defect
Area (expressed in terms of square feet)	Not as marked on the bend.
Thickness 1/	Not as specified.
Marking	Omitted, illegible or not in area specified.

1/ Thickness shall be determined in accordance with method 1011 of of FED-STD-311, except that no individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.2.3 Examination of packaging requirements. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that the packaging, packing and marking comply with the requirements of section 5.

* 4.3 End item testing. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table II, shall be followed. Sampling procedures and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test results shall contain the individual values utilized in expressing final result.

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TABLE II. Test Methods.

Characteristics	Requirement paragraph	Test method
Finish	3.4.3	1/
Corrosive effects of leather	3.4.4	3203 2/
Resistance to low temperature	3.4.5	4.3.1
Resistance to high temperature	3.4.6	4.3.2
Tensile strength	3.4.7	2021 3/
Total ash	3.4.8	6421
Water soluble	3.4.8	6341

1/ The contractor shall retain in his possession, and have available for inspection by the Government representative, documental evidence showing that the item conforms to the requirement for finish. The Government reserves the right to check test the item to verify its conformance.

2/ The following metals shall be used:

Specification	Material
QQ-A-250/4	Aluminum
QQ-B-626	Brass
QQ-B-750	Bronze
MIL-S-7720	Stainless steel, condition A
MIL-S-18729	Steel

3/ The number of tests in each direction shall be 15.

* 4.3.1 Resistance to low temperatures. The resistance to low temperature shall be determined as specified in Method 7211 of FED-STD-311, except that the specimen shall be exposed to a temperature of -56.7° ± 1.1°C (-70°F ± 2°F) for 24 hours.

4.3.2 Resistance to high temperature. The specimen for test shall be approximately 2 by 2 inches cut from the sample unit of leather. It shall be folded crosswise sharply upon itself with the grain side out and shall be exposed to a temperature of 126.7° ± 1.1°C (260° ± 2°F) for 18 hours. At the end of this period the specimen shall be examined for cracks in the area around the bend.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. The leather is intended for use as gaskets, padding, seals, and retainers.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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